

Electromagnetic Waves

SHORT NOTES

❖ Four Maxwell's Equations

$$1. \oint \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0}$$

$$2. \oint \vec{B} \cdot d\vec{s} = 0$$

$$3. \oint \vec{E} \cdot d\vec{l} = -\frac{d}{dt} \phi_B = -\frac{d}{dt} \int \vec{B} \cdot d\vec{s}$$

$$4. \oint \vec{B} \cdot d\vec{l} = \mu_0 (I_C + I_D) = \mu_0 \left(I_C + \epsilon_0 \frac{d\phi_E}{dt} \right)$$

$$\text{❖ Displacement current } I_D = \epsilon_0 \frac{d}{dt} \phi_E = \epsilon_0 \frac{d \oint \vec{E} \cdot d\vec{s}}{dt} = C \frac{dV}{dt}$$

❖ The electric and magnetic fields wave equations for an EM wave.

$$\frac{\partial^2 E}{\partial x^2} = \mu_0 \epsilon_0 \frac{\partial^2 E}{\partial t^2}; \frac{\partial^2 B}{\partial x^2} = \mu_0 \epsilon_0 \frac{\partial^2 B}{\partial t^2}$$

$$+ E = E_0 \sin(\omega t - kx) \text{ and } B = B_0 \sin(\omega t - kx)$$

$$+ c_{\text{vacuum}} = \frac{1}{\sqrt{\mu_0 \epsilon_0}}; V_{\text{medium}} = \frac{1}{\sqrt{\mu_r \mu_0 \epsilon_r \epsilon_0}}$$

$$+ \text{Refractive index of medium, } n = \sqrt{\mu_r \epsilon_r}$$

$$+ \frac{E_0}{B_0} = \frac{E_{\text{RMS}}}{B_{\text{RMS}}} = \frac{E}{B} = c$$

$$\text{❖ Intensity, } I = \frac{\text{power (P)}}{\text{Area (A)}}$$

$$\text{❖ Average intensity of wave } I_{\text{av}} = (\text{average energy density}) \times (\text{speed of light}), I_{\text{av}} = U_{\text{av}} \cdot c = \frac{1}{2} \epsilon_0 E_{\text{rms}}^2 c = \frac{2B_{\text{rms}}^2}{\mu_0} c$$

$$\text{❖ Instantaneous energy density, } u = u_E = u_B$$

$$u = \frac{1}{2} \epsilon_0 E^2 = \frac{B^2}{2\mu_0}$$

$$\text{❖ Average energy density } u_{\text{av}} = \frac{1}{4} \epsilon_0 E_0^2 = \frac{B_0^2}{4\mu_0} = \frac{\epsilon_0 E_{\text{rms}}^2}{2} = \frac{B_{\text{rms}}^2}{2\mu_0}$$

$$\text{❖ Energy = (momentum). } c \text{ or } U = Pc$$

$$\text{❖ Radiation pressure} = \frac{\text{Intensity}}{c} \text{ (when the wave is completely absorbed)}$$

$$= \frac{2(\text{Intensity})}{c} \text{ (when the wave is completely reflected)}$$

$$\text{❖ Intensity of wave from a source at a distance } r \text{ from it is proportional to } \frac{1}{r^2} \text{ (for a point source)}$$

$$\frac{1}{r} \text{ (for a line source)}$$

For a plane source intensity is constant & independent of r.